

Unit 5: Correlation

Univariate data- The data involving only one variable is called univariate data e.g. let variable under consideration is height of students, if we collect height of 'n' students, we have $X_1, X_2, X_3, \dots, X_n$ observations, where X_i ($i = 1, 2, 3, \dots, n$) denotes height of i^{th} student. Thus the set of observations on single variable is a univariate data.

Bivariate data- The data involving two variable is called bivariate data e.g. let variables under consideration are height (X) and weight (Y) of students, if we collect height and weight of 'n' students, we have $(X_1, Y_1), (X_2, Y_2), (X_3, Y_3), \dots, (X_n, Y_n)$ observations, where (X_i, Y_i) ($i = 1, 2, 3, \dots, n$) denotes height and weight of i^{th} student. Thus the set of observations on two variables is a bivariate data.

Concept of correlation-

Some times we come across the situations in which we are interested in observations on two variables for every unit in the sample (i.e. bivariate data). Consider the following examples

Height and Weight of students, Income and Expenditure of families, Demand and Supply of a commodities, Petrol consumption and Distance travelled by car, Age of applicant and LIC premium, Demand and price of a commodity, Development in medical science and Death rate etc.

In all above examples it is observed that as one variable changes other also changes in same or opposite direction. This implies that there is some relation between two variables, this relation is called correlation and corresponding two variables are said to be correlated.

Defⁿ- The study of existence of magnitude and direction of relation between two variables is called correlation. OR The extent of linear relationship between two variables is called correlation.

Types of correlation-

1) Positive and Negative correlation-

This type of correlation is based on direction of change in two variables.

If both the variables are changes in same direction i.e. as one variable increases other also increases or as one variable decreases other also decreases, then there is positive correlation between two variables. E.g. Income and Expenditure of families, Expenditure on advertisement and sale of commodity, Age of applicant and LIC premium etc.

If both the variables are changes in opposite direction i.e. as one variable increases other is decreases or as one variable decreases other is increases, then there is negative correlation between two variables. E.g. Price and Demand of a commodity, Development in medical science and Death rate, T.V. registration and Attendance at theatre etc.

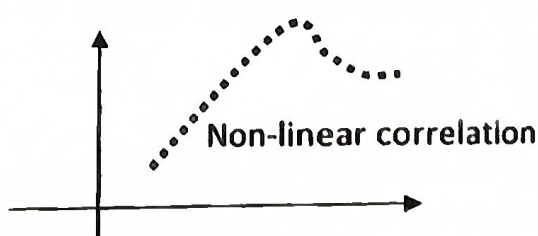
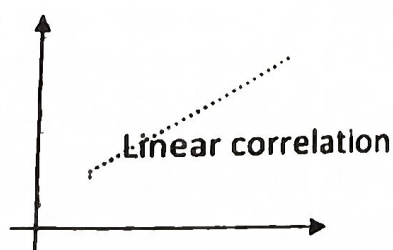
If change in one variable is proportional to change in other variable then there is perfect correlation, moreover if change is in same direction then correlation is perfect positive and if change is in opposite direction then correlation is perfect negative.

2) Linear and Non-linear correlation-

This type of correlation is based on the nature of the graph of the relation between two variables.

If the graph is a straight line the correlation is linear e.g. built up area and cement used, petrol consumption and distance travelled by motorcycle.

If the graph is not a straight line but a curve the correlation is Non-linear e.g. Rain fall & production of pulses, quantity of fertilizer used & production of sugarcane.



Correlation & Casual relation:- When two variables are correlated one of them is cause and other is effect . e.g. We know that Rainfall and Production of rice are correlated, we also know that production of rice is the effect and rainfall is the cause. Similarly Expenditure on advertisement and Sale of commodity are correlated, in this case sale of commodity is the effect and expenditure on advertisement is the cause.

Thus we expect the correlation when we know there is casual relation and conversely we may expect casual relation when we know there is correlation. But it should be noted carefully that, it is not true always, in some cases we may find correlation and yet there may not be any casual relation. Such a false correlation is called non-sense correlation. This happens because of following reasons.

i) The correlation may be due to pure chance we may get high degree correlation from the figures obtained and in reality there is no causal relation. This happens mainly due to faulty sampling and bias of investigator. E.g. A sample of weight and marks of students may show high correlation, but we know that weight and marks are not related. This is non-sense correlation and due to pure chance or coincidence.

ii) Both the variables may be depends upon third variable- It may happen that both the variables are depends upon third variable. In such cases we may get high degree correlation but this does not imply a causal relation. E.g. Number of cars and T.V. sets in different cities may show correlation, but we know that one is not a cause of other. This happens because both are the effect of rise in living standard and rise in purchasing power of people. Similarly production of rice and production of tea both are dependence on rainfall.

Thus while interpreting correlation it is necessary to be careful, otherwise we may reach at a wrong conclusion.

Methods of studying correlation-

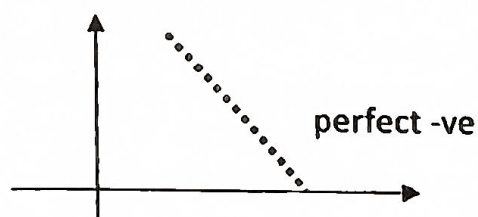
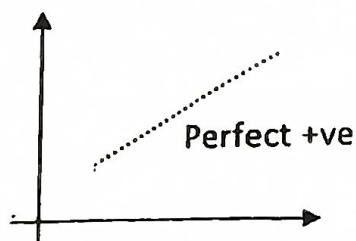
- 1) Scatter diagram method
- 2) Karl Pearson's coefficient of correlation

1) Scatter diagram method-

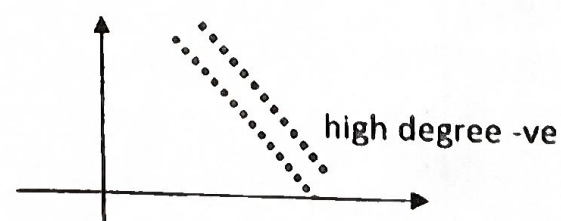
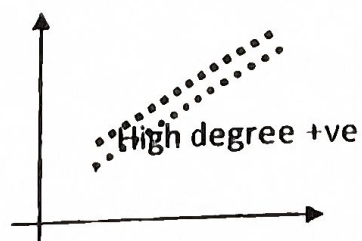
Scatter diagram is a graph showing the correlation between two variables. To obtain scatter diagram, the values of two variables are plotted on graph paper, taking one variable on X-axis and other on Y-axis. by plotting data, we get points which are generally scattered, so this method is known as scatter diagram. The way in which points are scattered indicates the degree and direction of correlation between two variables.

Interpretation of scatter diagram-

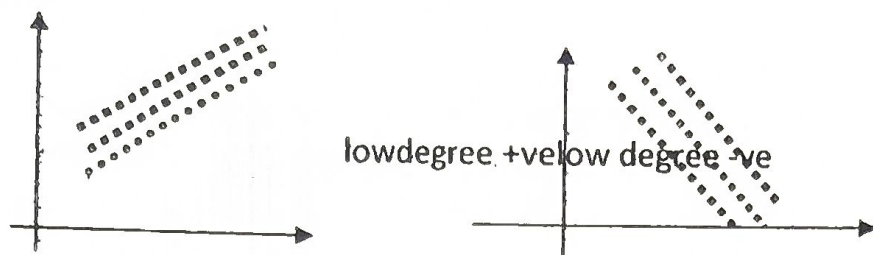
i) If the points are lie on a straight line which rising (falling) to wards right, then the correlation is perfect +ve(-ve)



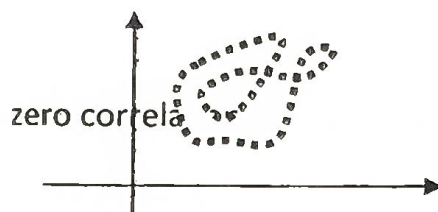
ii) If the points are lie in a narrow strip which is rising (falling) to wards right, then the correlation is high degree +ve(-ve)



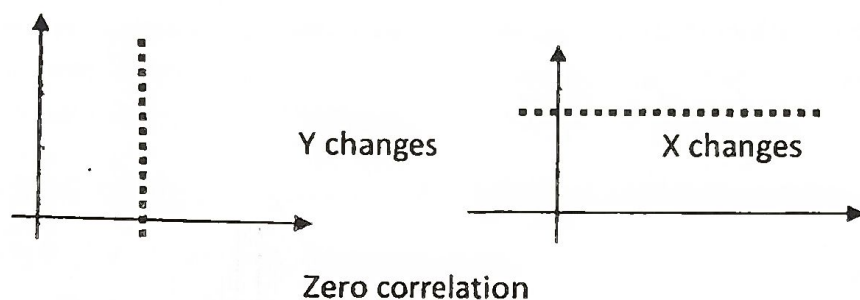
iii) If the points are lie in a broad strip which is rising (falling) to wards right, then the correlation is low degree +ve(-ve)



iv) If the points are scattered at random without any pattern, then there is zero correlation (no correlation) between two variable



v) If the points are lie on a straight line which is parallel to X-axis or Y-axis (i.e. one variable is changes and other is constant), then there is no or zero correlation.



Merits and Demerits of scatter diagram method-

Merits-

- i) It is simplest method of studying correlation.
- ii) It is easy to understand.
- iii) It is less affected by extreme observations.

Demrrits-

- i) It does not gives numerical measure of correlation.
- ii) It gives rough idea about correlation.
- iii) It is not suitable for large data.

Covariance – Let (X_i, Y_i) , $i = 1, 2, 3, \dots, n$ be the bivariate data on (X, Y) . the covariance between two variables X and Y is denoted by $\text{cov}(X, Y)$ and defined as

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

Note- i) for computational purpose formula can be simplified a

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n} = \left(\frac{\sum XY}{n} \right) - \bar{X}\bar{Y}$$

$$\text{ii) } \text{cov}(X, Y) = \text{Cov}(Y, X)$$

iii) value of $\text{Cov}(X, Y)$ may be +ve or -ve

iv) $\text{cov}(X, Y)$ can be considered as joint central moment of order (1,1) of (X, Y) i.e. $\mu_{(1,1)} = \text{Cov}(X, Y)$

v) $\text{Cov}(X, A) = 0$ where A is constant

Remark- The correlation between two variables X & Y is nothing but the study of joint or simultaneous variation between the variables X & Y. $Cov(X, Y)$ measures the joint variation between the variables X & Y, its value may be +ve or -ve according to correlation is +ve or -ve. Therefore Karl Pearson used covariance to define correlation.

Karl Pearson's coefficient of correlation-Defⁿ- The Karl Pearson's coefficient of correlation between two variables X & Y is denoted by 'r' or ' r_{xy} ' or $corr(X, Y)$ and defined as

$$r = \frac{Cov(X, Y)}{\sigma_x \cdot \sigma_y}$$

where σ_x and σ_y are s.d. of X & Y respectively.

Note -

i) $r_{xy} = \frac{\mu_{11}}{\sqrt{\mu_2(x) \cdot \mu_2(y)}}$, So Karl Pearson's coefficient of correlation is also known as product moment

coefficient of correlation.

ii) $r_{xy} = r_{yx}$

iii) $corr(X, A) = 0$, where A is constant

iv) For computational purpose the formula of correlation can be simplified as

$$r = \frac{Cov(X, Y)}{\sigma_x \cdot \sigma_y} \text{ or } = \frac{\frac{1}{n} \sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\frac{1}{n} \sum (X - \bar{X})^2 \cdot \frac{1}{n} \sum (Y - \bar{Y})^2}}$$

$$\text{or } = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \cdot \sum (Y - \bar{Y})^2}}$$

$$\text{or } = \frac{\sum XY - n\bar{X}\bar{Y}}{\sqrt{(\sum X^2 - n(\bar{X})^2)(\sum Y^2 - n(\bar{Y})^2)}}$$

$$\text{or } = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{n}}{\sqrt{\left(\sum X^2 - \frac{(\sum X)^2}{n}\right)\left(\sum Y^2 - \frac{(\sum Y)^2}{n}\right)}}$$

#Properties of Karl Pearson's coefficient of correlation-

1) Karl Pearson's coefficient of correlation is independent of change of origin and scale

i.e. let $(X_i, Y_i), i = 1, 2, 3, \dots, n$ be a set of pairs of n observations on bivariate (X, Y)

By definition $r_{xy} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \cdot \sum (Y - \bar{Y})^2}}$

Let $u = \frac{(X - A)}{B}$ and $v = \frac{(Y - C)}{D}$ where A, B, C, D are the constants.

By definition $r_{uv} = \frac{\sum (u - \bar{u})(v - \bar{v})}{\sqrt{\sum (u - \bar{u})^2 \cdot \sum (v - \bar{v})^2}}$

Then above property implies that $r_{xy} = r_{uv}$

2) The Karl Pearson's coefficient of correlation is always lies between -1 and 1 i.e. $-1 \leq r \leq 1$

Interpretation of Karl Pearson's coefficient of correlation.

The numerical value of correlation coefficient measures the degree of relationship between two variables and sign shows direction of change.

- i) If $r = +1$, then there is perfect +ve correlation between two variables
- ii) If $r = -1$, then there is perfect -ve correlation between two variables
- iii) If $r = 0$, then there no correlation between two variables

The assumptions on which Karl Pearson's coefficient of correlation is based-

- i) There is linear relationship between two variables i.e. scatter diagram will give straight line only.
- ii) There is cause and effect relationship between two variables.

Merits and Demerits of Karl Pearson's coefficient of correlation

Merits-

- i) It gives numerical value and direction of correlation.
- ii) It is useful for estimation
- iii) It is based on all observations.

Demerits-

- i) It is affected by extreme observations.
- ii) It measures only linear relationship.
- iii) It is not suitable for qualitative data

Ex1. Find correlation coeff. between height and weight of a group of 6 students.

Height (cm)	165	155	157	171	160
Weight (kg)	60	62	58	72	57

Ex2. Find correlation coeff. between demand and supply of a commodities.

demand (kg)	65	55	57	71	60
supply (kg)	52	62	58	51	57

Ex3. Find correlation coeff. Between X and Y if

$n = 10, \sum X = 164, \sum Y = 225, \sum X^2 = 3126, \sum Y^2 = 6149, \sum XY = 4237$

Ex4. Find correlation coeff. Between X and Y if

$$n = 6, \sum(X - 18.5) = -3, \sum(Y - 50) = 20, \sum(X - 18.5)^2 = 19, \sum(Y - 50)^2 = 850, \sum(X - 18.5)(Y - 50) = -120$$

Ex5. Find n If correlation coeff. Between X and Y is 0.5, $\sum(X - \bar{X})(Y - \bar{Y}) = 60$, $\sum(X - \bar{X})^2 = 90$, $\sigma_y = 4$

Ex6. Given : $n = 10$, $\bar{X} = 10$, $\bar{Y} = 6$, $\sum XY = 650$, $\sigma_x = 3$, $\sigma_y = 2$ Find 'r'

Ex1. Calculate the Karl's Spearman's correlation coefficient between the marks given by two judges to 8 competitors in the competition of one act play.

Marks by judge -I	81	72	60	33	29	11	56	42
Marks by judge-II	75	56	42	15	30	20	60	80

Ex2. The runs record by 11 cricketers in two tests matches are as follows. Find Correlation coefficient and interpret the result.

Runs in match -I	82	82	85	35	55	33	52	53	55	50	56
Runs in match -II	50	50	93	31	66	37	87	50	62	64	65